CityPulse: A Multi-Sensor Urban Hazard Network

Designed for Karachi, built and evaluated on public data

Course: Machine Learning (Practical), AI-3202, BS Artificial Intelligence, DUET Karachi

Deliverable: Lab 14, Complex Computing Activity (CCA)

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Status: Proposal v2.0 (public-data-only, 2-week plan)

Revision log

Version	Change
v1.0	Initial merged proposal (SafeRoute + EchoMap)
v2.0	**All self-collected and synthetic data removed.** Every dataset is now publicly available. Karachi-specific claims removed; the system is city-agnostic and evaluated on public data. Real geo-tagged public complaint data (NYC 311) replaces the synthetic report layer. Generator sound class and the phone hold-out set dropped (no public data). Cross-dataset testing added.

1. Summary

CityPulse is a web application where a user submits a **photo** of a road hazard or a **5-second sound clip** of a noise source, tagged with a location and timestamp. Machine learning then:

1. classifies the photo or sound,
2. scores how trustworthy the report is,
3. merges duplicate reports into single incidents,
4. discovers chronic hotspot zones,

5. predicts future risk (flooding, noise peaks), and
6. shows everything on one live **livability map** with a **"Calm & Safe Route"** planner.

The design targets cities like Karachi, where authorities lack live location-level visibility into road hazards and noise. **We do not collect or create any data ourselves.** Everything is trained and evaluated on public datasets, and the live map is demonstrated on a **public city complaint dataset (NYC 311)** that already contains geo-tagged, time-stamped hazard and noise reports. The architecture is city-agnostic, so a Karachi deployment would only need a local data source.

Sensor	Input	Classifies	Example output
Eyes	Photo	Pothole, road crack, flooded road, normal road	"Pothole, 92% confidence"
Ears	5 s audio	Traffic/horns, construction, siren, music/loudspeaker, HVAC/mechanical	"Construction, relative intensity: high"

2. Problem Statement

Cities such as Karachi face road hazards (potholes, waterlogged underpasses) and chronic noise pollution, but complaints arrive unstructured, unverified, and duplicated. Authorities cannot easily see where problems cluster, which reports are genuine, or where risk will rise next (for example, after heavy rain).

Goal: show that crowd-sourced reports can be turned into a verified, deduplicated, ranked, and predictive hazard picture using classical machine learning, and demonstrate the full pipeline end to end on public data.

3. Objectives and Non-Goals

Objectives

1. Build an end-to-end ML system (preprocessing, modeling, evaluation, serving, visualization) integrating at least three lab concepts. We commit to seven core labs and add baselines if time allows.
2. Classify road-hazard photos and urban-noise audio using public datasets.
3. Estimate report trustworthiness, merge duplicates, and discover hotspot zones on real public geo-tagged complaint data.

4. Forecast risk from weather and time features.
5. Deliver a working full-stack demo: FastAPI backend plus vanilla HTML/CSS/JS frontend.

Non-goals

- Collecting any new data (photos, audio, GPS, surveys). Not allowed by the project rules we adopted.
- Generating synthetic data.
- Reporting legal decibel (dB) levels. Public clips are not calibrated, so we report source type and **relative** intensity only.
- Claiming measured performance on Karachi or on real budget-phone recordings. We have no such data, and the report says so.

4. Data Policy

1. **Public data only.** Every dataset is downloaded from a public source with a documented license (Section 7).
2. **No synthetic data.** Every number in the report comes from real public data.
3. **Honest scope.** Results describe the public datasets used (mostly non-Karachi). The report does not generalize them to Karachi.
4. **Demo mode.** During the live demo, "submitted" photos and clips are samples taken from the **held-out test split** of public datasets, placed on the map by clicking a location. These demo submissions are not stored as new data, are never used for training, and are not reported as results.

5. How It Works (Pipeline)

1. **Input (vanilla JS).** The page lets a user pick a photo or audio sample (demo mode) or use the device camera/mic, and place a pin on a Leaflet map. Timestamp is attached automatically.
2. **Preprocessing (Lab 1).** Images are resized and normalized. Audio is converted to a fixed-length Mel-spectrogram (Librosa). Complaint records are cleaned (missing coordinates, outliers, time features).
3. **Classification (Labs 3, 4, 5, 6, 8, 9).** MobileNetV2 embeddings (images) and a small CNN (spectrograms) feed SVM and Random Forest, compared with KNN, Naive Bayes, and Decision Tree baselines.

4. **Trust scoring (Lab 3).** Logistic regression predicts whether a complaint is likely valid (Section 6, row Lab 3).
5. **Duplicate merging (Lab 11).** Hierarchical clustering merges reports of the same hazard into one incident.
6. **Zone discovery (Lab 10).** K-Means with the elbow method and silhouette finds chronic hotspot zones.
7. **Prediction (Lab 2).** Regression forecasts street-flooding complaints from rainfall and noise complaints from time-of-day features.
8. **Serve.** FastAPI endpoints return incidents, zones, predictions, and routes.
9. **Visualize.** Leaflet dashboard with layers (road hazards, noise, combined livability), a time slider, and a municipal repair-priority view.

```
[Browser: HTML/CSS/JS + Leaflet + Chart.js]
        | fetch()
[FastAPI backend]
   |-- POST /report      classify photo/audio, trust-score
   |-- GET  /incidents   deduplicated incidents (GeoJSON)
   |-- GET  /zones       K-Means hotspot zones + livability score
   |-- POST /predict     flood-risk / noise-peak forecast
   |-- POST /route       risk-weighted "Calm & Safe" route
   |-- GET  /admin       ranked repair-priority list
        |
[SQLite: loaded public 311 subset]  +  [Saved models (joblib / CNN weights)]
```

5.1 Shared report schema

```
{
  "id": "string",
  "modality": "image | audio | complaint",
  "type": "pothole | crack | flood | traffic_horn | construction | siren |
music_loudspeaker | hvac | normal",
  "confidence": 0.92,
  "relative_intensity": 0.0,
  "lat": 40.7128,
  "lon": -74.0060,
  "timestamp": "2024-06-15T14:30:00",
  "source": "nyc311 | demo",
  "trust_score": 0.88,
  "incident_id": "string-or-null",
  "zone_id": "int-or-null"
}
```

`modality: complaint` rows come from the public 311 data. `source: demo` marks demo-mode submissions drawn from public test splits.

5.2 Key features

- **Livability Score** per neighbourhood, fusing road damage, flood risk, and noise.
- **"Calm & Safe Route" mode:** routes avoid high-hazard clusters and chronic noise zones (demo centerpiece).
- **Rain mode:** enter a rainfall amount and the map highlights zones with elevated flood-complaint risk, with a **report-density overlay** showing where data is thin.
- **Explainability panel:** click a zone to see why it is rated critical (feature importance, contributing reports).
- **Municipal view:** repair priority ranked by severity and number of reports.

6. Lab Concept Map

Lab	Concept	Where it is used	Data it runs on	Problem it solves	Evidence for the report
1	Cleaning, encoding, scaling, splitting	Image resize/normalize; audio to fixed-length Mel-spectrogram; dropping 311 rows with missing coordinates; timestamp to hour/weekday; one-hot encoding of complaint type and borough; StandardScaler on tabular features	All datasets	Raw data differs in format, length, resolution, and completeness	Before/after row counts, missing-value table
3	Logistic regression	**Trust scorer.** Predicts whether a complaint was found valid. Label is a **proxy** derived from the 311 resolution outcome (action taken vs unable to locate / no evidence). Features: complaint type and descriptor, hour, weekday, submission channel, location type, local complaint density	NYC 311	Invalid or low-value reports waste response effort	Precision/recall, ROC/PR curve, confusion matrix; the report states the label is a proxy
6	SVM + GridSearchCV	Main classifier on CNN embeddings (images and spectrograms); tune C, kernel, gamma	UrbanSound8K, ESC-50, RDD2022, flood image sets	High-dimensional embeddings, limited labels	Best params, per-class F1

Lab	Concept	Where it is used	Data it runs on	Problem it solves	Evidence for the report
5	Random Forest	Second classifier plus feature importance for the explainability panel	Same as above, plus 311 for zone-risk features	Compare with SVM; explain why a zone is risky	RF vs SVM table, importance chart
4	Decision Tree	Baseline classifier and readable rules for the municipal view	Same	Officials need interpretable rules	Tree plot, depth vs accuracy (overfitting study)
8	KNN	Baseline classifier; nearest-neighbour lookup of similar past incidents	Same	Simple comparison model	Accuracy vs K curve
9	Naive Bayes	Baseline on tabular complaint metadata (hour, borough, descriptor)	NYC 311	Cheap probabilistic baseline	Accuracy and calibration vs other models
10	K-Means	Geo-clustering of complaint coordinates into hotspot zones (elbow + silhouette)	NYC 311	Thousands of points become a few actionable zones	Elbow plot, silhouette score, zone map
11	Hierarchical agglomerative clustering	**Duplicate merging** within distance and time thresholds; dendrogram cut selects the threshold	NYC 311	Many reports of one hazard should count as one incident	Dendrogram, cophenetic correlation, ARI against **weak labels** (same address + same complaint type within a short window)
2	Linear / multiple regression	(a) daily street-flooding complaint count from rainfall and lagged rainfall; (b) noise-complaint volume from hour, weekday, borough	NYC 311 + Open-Meteo	Forecast when and where problems peak	MSE, RMSE, R^2, predicted-vs-actual and residual plots (time-based split)

Lab	Concept	Where it is used	Data it runs on	Problem it solves	Evidence for the report
12 (optional)	ML overlay on live camera	Camera overlay of hazard labels and confidence	Public test images	Wow-factor demo, only if time allows	Latency (ms), FPS

Priority. Core: Labs 1, 2, 3, 5, 6, 10, 11. With a 2-week deadline, Labs 4, 8, 9 (baselines) are built only if the core is done by Day 9, and Lab 12 (AR) is dropped.

7. Datasets (all public)

> **Verification note.** Dataset facts below were checked against their public pages or papers. Complaint-type names, label columns, and download links must be re-confirmed by the team on the day they download (field names can change).

7.1 Audio ("ears")

UrbanSound8K (main training set)

- 8,732 labeled urban sound excerpts (up to 4 s each) in 10 classes: air conditioner, car horn, children playing, dog bark, drilling, engine idling, gun shot, jackhammer, siren, street music.
- Pre-sorted into 10 folds. The authors advise using the predefined folds with 10-fold cross-validation, not reshuffling, to avoid leakage between clips from the same recording. **We follow this.**
- Download: Zenodo record 1203745, Hugging Face danavery/urbansound8K, or Kaggle.
- License: Creative Commons Attribution-NonCommercial (fine for coursework). Cite Salamon, Jacoby and Bello (2014).

ESC-50 (supplementary and cross-dataset test)

- 2,000 environmental recordings: 50 classes, 40 five-second clips each, 5 folds. The 5 s length matches the app's clip length.
- Relevant classes include chainsaw, hand saw, siren, car horn, and engine.
- License: CC BY-NC (the 10-class ESC-10 subset is CC BY). Caveat from the maintainers: some source recordings were pre-processed (mostly band-limited) in a class-dependent way, which can leak information. We note this under limitations.
- Download: https://github.com/karolpiczak/ESC-50. Cite Piczak (2015).

Final audio classes (public data only)

App class	Source clips
traffic_horn	UrbanSound8K car_horn, engine_idling; ESC-50 car horn, engine
construction	UrbanSound8K drilling, jackhammer; ESC-50 chainsaw, hand saw
siren	UrbanSound8K siren; ESC-50 siren
music_loudspeaker	UrbanSound8K street_music
hvac	UrbanSound8K air_conditioner
(excluded)	gun_shot (sensitive); dog_bark, children_playing optionally as a background class

The earlier "generator" class is **dropped**: no public dataset for it was found, and we cannot record our own.

Phone-microphone realism without collecting data. We cannot test on real phone recordings. Instead we (a) apply **augmentations to the public training clips** that mimic cheap microphones (band-limiting, compression, added background noise), and (b) run **cross-dataset testing**: train on UrbanSound8K, test on ESC-50 clips of matching classes. The report calls (a) "simulated degradation", not real phone performance.

7.2 Images ("eyes")

RDD2022 (main road-damage set)

- 47,420 road images from six countries (Japan, India, Czech Republic, Norway, US, China) with more than 55,000 annotated damage instances in four types: longitudinal crack (D00), transverse crack (D10), alligator crack (D20), pothole (D40).
- **The India subset (about 9,665 images) is the most relevant.** India, Japan, and Czech images were captured with smartphones mounted on vehicles.
- Format: JPG plus Pascal VOC XML bounding boxes. Annotations are provided for the training set only, so we make our own train/validation/test split.
- Labels are detection boxes. For classification we crop box regions into patches or derive image-level labels (pothole / crack). For the normal road class, confirm that undamaged images exist in the release; otherwise crop patches away from annotated

boxes.

- License: CC BY-SA 4.0. Size: about 13 GB.
- Download: https://github.com/sekilab/RoadDamageDetector (links to the Figshare release). Cite Arya et al., RDD2022.

Flood images (ground-level)

Dataset	What it is	Size (as reported)	Use and caveats
AlleyFloodNet (Lee and Joo, 2025)	Ground-level flood vs non-flood images from alleys and lowlands	Train 454 flood / 322 non-flood; validation 92 / 63; test 114 / 81	Best match for phone-style views. The paper is a **preprint (not peer-reviewed)**; confirm the dataset download link and license in the paper
Floodwater-on-roadways dataset (arXiv 1909.00125)	Hand-labeled roadway flood images from varied locations, released by the authors	253 flood and 238 non-flood images	Small but road-focused; used for classification and as an extra test set
MediaEval 2018 flood passability	Social-media images labeled for road passability	7,387 development and 3,683 test images	Large, but verify availability and the Flickr image licenses before use
FloodNet (Rahnemoonfar et al., 2021)	**Aerial (UAV)** images after Hurricane Harvey	2,343 images in the supervised release	Aerial, not street-level. Use only as a minor supplement; a model trained mostly on it will not transfer to phone-style photos

We combine the ground-level sets for the flood class. Their small size is a stated limitation, so flood classification results are reported with confidence intervals and not over-claimed.

7.3 Geo-tagged complaint data (Labs 2, 3, 9, 10, 11)

NYC 311 Service Requests (NYC Open Data)

- Public city complaint records with creation and closing timestamps, complaint type, descriptor, status, resolution text, submission channel, and latitude/longitude.
- Two releases: **2010 to 2019** and **2020 to present** (the latter is updated regularly). The data does not expose personally identifying information about the requester.
- Access: the Socrata (SODA) API at data.cityofnewyork.us, or CSV download. Dataset pages: 2020-present id erm2-nwe9, 2010-2019 id 76ig-c548.
- Why it fits: its noise records carry descriptors such as car/truck horn, engine idling, and car/truck music, which map naturally onto our audio classes. It also contains street-condition (pothole) and street-flooding-related requests (public analyses of NYC street-flooding complaints exist). **The team must query the distinct complaint_type and descriptor values and confirm the exact names** before building filters.
- Scope for manageability: restrict to **one borough and one to two years** (millions of rows overall). Keep only rows with valid coordinates.
- Uses: K-Means zones (Lab 10), duplicate merging (Lab 11), trust scoring (Lab 3), Naive Bayes baseline (Lab 9), noise and flooding regression (Lab 2).

Trust-score label (proxy, must be stated as such). Resolution outcomes such as "unable to locate", "no evidence", or "duplicate" are treated as **invalid**; outcomes where an agency acted or a violation was found are **valid**. Confirm the actual resolution text categories after loading, and document the mapping exactly. This label measures "did the city act on it", which is related to but not identical to "was it genuine".

Duplicate-merging evaluation (weak labels). There are no true duplicate labels. Use requests with the same address and complaint type within a short window (for example 24 hours) as **weak ground truth** for ARI, plus internal metrics (cophenetic correlation, silhouette).

7.4 Weather and map data

Open-Meteo Historical Weather API (rainfall for Lab 2)

- Historical weather from 1940, no API key, free for non-commercial use up to 10,000 calls per day. Data licensed CC BY 4.0 (attribution required).

- Endpoint: https://archive-api.open-meteo.com/v1/archive with the study city's coordinates and the hourly precipitation variable. We aggregate to daily totals and lagged totals for the flood-complaint regression.

OpenStreetMap (details to be verified by the team)

- Road network for the study area, used by osmnx and a routing engine (for example OSRM) for the "Calm & Safe Route" feature. License: ODbL (attribution required).

7.5 Optional Karachi context layer (no ML claims)

To keep the Karachi motivation visible without Karachi data we do not have, the dashboard may include a **Karachi view** built only from public sources: the OpenStreetMap road network and Open-Meteo rainfall history for Karachi. It is a **display-only context layer** (for example, historical rainfall and road network); no classifier, trust score, or forecast is claimed for Karachi.

7.6 Data-to-task summary and splits

Task	Dataset	Split rule
Audio classification	UrbanSound8K (+ ESC-50 subset)	Provided folds for 10-fold CV
Cross-dataset audio test	Train UrbanSound8K, test ESC-50 matching classes	Strict: no ESC-50 clip in training
Road-damage classification	RDD2022 (India first)	By image, stratified
Flood classification	AlleyFloodNet + floodwater-on-roadways (+ MediaEval if usable)	Use provided splits; one set kept as a pure test set
Trust scoring, zones, duplicates, Naive Bayes	NYC 311 subset	Time-based split for supervised models
Rainfall-based regression	NYC 311 + Open-Meteo	**Time-based split** (train earlier, test later)

8. Evaluation Plan

Component	Metrics
Audio and image classifiers	Accuracy, per-class precision/recall/F1, confusion matrix; SVM vs RF vs baselines; **in-dataset vs cross-dataset scores** to quantify domain shift; scores with and without simulated degradation
Trust scorer	Precision/recall, ROC/PR curves, calibration; reported as a proxy-label result
Duplicate merging	Dendrogram cut analysis, cophenetic correlation, ARI against weak labels
Hotspot zones	Elbow plot, silhouette score, Davies-Bouldin
Regression	MSE, RMSE, R^2, residual plot (time-based split)
System	API latency, time from submit to map update

9. Demo Plan

1. Load the dashboard showing real NYC 311 hazard and noise zones (K-Means hotspots).
2. In demo mode, submit a held-out public audio clip (for example, a jackhammer sample) at a map location. The app classifies it and drops a pin.
3. Submit a held-out public pothole image. A second pin appears.
4. Submit several reports near the same spot and watch them merge into one incident.
5. Switch on **Rain mode** and show the elevated-risk zones.
6. Click "Plan route": the path detours around the new hazards and the chronic noise zone.

Demo submissions are labeled source: demo and are not counted as data or results.

10. Risks and Mitigations

Risk	Mitigation
Domain shift (public, mostly non-Karachi data)	Cross-dataset tests; state plainly that results are not Karachi results
No public generator or Karachi-specific audio	Class set limited to what public data supports
Small ground-level flood datasets	Combine several sets; report confidence intervals; treat as a limitation
Proxy trust label	Document the exact mapping; call the result "complaint validity proxy"
No true duplicate labels	Weak labels plus internal clustering metrics; say so
Class imbalance (RDD2022)	Class weights or balanced sampling
Dataset size (RDD2022 about 13 GB; 311 has millions of rows)	Use the India subset; restrict 311 to one borough and a short window
Licensing	UrbanSound8K and ESC-50 are non-commercial; RDD2022 is CC BY-SA; check each flood set's license; attribute Open-Meteo and OSM
Link or field-name drift	Re-verify downloads and 311 column values on the day of download
Scope creep	Core labs first; baselines, Karachi view, and AR are stretch goals

11. Ethics and Fairness

- **No personal data is collected by us.** The 311 data is published without requester identities; image and audio sets are public research datasets. Demo-mode inputs come from those sets.
- **Reporting bias:** complaint data over-represents areas where people report more. Zone rankings should be normalized by reporting rate, and low-data areas flagged, not scored as safe. The 311 "validity" proxy also reflects agency behavior, which can differ

by neighbourhood.

- **Misuse:** relative noise intensity is not evidence of legal violations. The system supports awareness and prioritization, not enforcement.
- **Transparency:** the report lists every proxy label, weak label, and limitation.

12. Team Roles (4 students)

Role	Responsibilities
1. Audio lead	UrbanSound8K/ESC-50 pipeline, Mel-spectrograms, degradation augmentation, spectrogram CNN, cross-dataset evaluation
2. Image lead	RDD2022 preprocessing, flood dataset integration, image classifier, SVM/RF comparison
3. Data and ML lead	NYC 311 extraction and cleaning, trust scorer, K-Means zones, hierarchical duplicate merging, regression with Open-Meteo, routing weights
4. Full-stack lead	FastAPI endpoints, SQLite schema, model serving, Leaflet/Chart.js dashboard, demo mode, optional AR overlay

All members contribute to the report, README, and demo. Replace roles with names once assigned.

13. Timeline (2 weeks)

Days	Work	Output
1 to 2	Finalize scope; download datasets (UrbanSound8K, ESC-50, RDD2022 India subset, flood sets, NYC 311 for one borough and 1 to 2 years, Open-Meteo rainfall); query 311 complaint/descriptor/resolution values; set up repo and report schema	Data inventory, confirmed label mappings

Days	Work	Output
3 to 5	Preprocessing (audio spectrograms, image patches, 311 cleaning); SVM and Random Forest classifiers on audio and images; FastAPI skeleton and dashboard shell in parallel	Cleaned datasets, first classifier metrics, running API stub
6 to 8	Trust scorer (Lab 3); K-Means zones (Lab 10); hierarchical duplicate merging (Lab 11); rainfall regression (Lab 2); connect endpoints to models	Model results, dendrogram, zone map, working endpoints
9 to 11	Dashboard map layers, demo mode, Calm & Safe Route; cross-dataset audio test; optional baselines (Decision Tree, KNN, Naive Bayes) only if core is done	Working end-to-end prototype, evaluation tables
12 to 13	Final evaluation and figures; 3 to 5 page report; README; record demo video	Report, README, demo video
14	Buffer, bug fixes, demo rehearsal, submission	Final repo and submission

2-week scope. Must have: Labs 1, 2, 3, 5, 6, 10, 11 and the full-stack demo. Image classes limited to pothole, crack, flooded road, and normal road; audio classes as in Section 7.1. Cut or only if ahead: Labs 4, 8, 9 baselines, Lab 12 AR overlay, the Karachi context view, and MediaEval and FloodNet flood data. Decide on Day 9 what to drop so the report and demo are never rushed.

14. Deliverables (mapped to CCA requirements)

- **Code:** all Python files/notebooks, inline comments, and a README.md with setup and run instructions.
- **Project report (3 to 5 pages):**
  1. Architecture and pipeline diagram
  2. Explicit list of lab concepts integrated and where (Section 6)
  3. Dataset description, licenses, and preprocessing steps (Section 7)
  4. Model designs, hyperparameters, training details, evaluation metrics (Section 8)
  5. Challenges, limitations, proxy labels, fairness/ethics (Sections 10 and 11)

6. Setup and run instructions (conda/pip environment, hardware needs, sample commands)

- **Presentation and demo:** live demo or recorded video.

Assessment alignment (CCA rubric, 10 marks): Design of CCA (2), Implementation and Execution (3), Problem Solving and Creativity (2), Documentation and Presentation (3).

15. Suggested Repository Structure

```
citypulse/
|-- README.md
|-- requirements.txt
|-- data/
|   |-- raw/            # UrbanSound8K, ESC-50, RDD2022 subset, flood sets,
311 extract
|   `-- processed/      # spectrograms, resized images, embeddings, cleaned
311
|-- notebooks/          # EDA, model comparison, clustering, regression
|-- src/
|   |-- preprocessing/  # audio.py, image.py, complaints.py
|   |-- models/         # classifiers.py, trust.py, regression.py
|   `-- clustering/     # kmeans_zones.py, hierarchical_dedup.py
|-- backend/            # FastAPI app, DB models, routes
|-- frontend/           # index.html, style.css, app.js (Leaflet, Chart.js)
|-- models/             # saved joblib / CNN weights
`-- report/             # final report, diagrams, figures
```

Do not commit raw datasets to Git; list download commands and license notes in the README instead.

16. References

1. J. Salamon, C. Jacoby, J. P. Bello. *A Dataset and Taxonomy for Urban Sound Research.* ACM Multimedia, 2014. (UrbanSound8K)
2. K. J. Piczak. *ESC: Dataset for Environmental Sound Classification.* ACM Multimedia, 2015. https://github.com/karolpiczak/ESC-50
3. D. Arya et al. *RDD2022: A multi-national image dataset for automatic road damage detection.* Geoscience Data Journal. https://github.com/sekilab/RoadDamageDetector

4. O. Lee, H. Joo. *AlleyFloodNet: A Ground-Level Image Dataset for Rapid Flood Detection in Economically and Flood-Vulnerable Areas.* Preprints.org, 2025 (preprint). https://www.preprints.org/manuscript/202504.0154
5. *Detecting floodwater on roadways from image data with handcrafted features and deep transfer learning.* arXiv:1909.00125.
6. M. Rahnemoonfar et al. *FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding.* IEEE Access, 2021. https://github.com/BinaLab/FloodNet-Supervised_v1.0
7. MediaEval 2018 Multimedia Satellite Task (flood passability benchmark), as described in the survey *Natural Disasters Detection in Social Media and Satellite Imagery* (arXiv:1901.04277).
8. NYC Open Data. *311 Service Requests from 2020 to Present* (erm2-nwe9) and *311 Service Requests from 2010 to 2019* (76ig-c548). https://data.cityofnewyork.us
9. Open-Meteo Historical Weather API. https://open-meteo.com (CC BY 4.0)
10. OpenStreetMap contributors (ODbL).